

M2Go1 - Automated Lemon Harvesting Robot

Smart Agriculture

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M2: Afifah Maisarah binti Asmadi

M3: Chew Yong Jun

M4: Yong Vin Cent





Questionnaire

1. How do you currently harvest fruits or crops from the trees?
2. What are the biggest challenge or difficulties you face during harvesting?
3. Which part of the harvesting process takes the most time or effort?
4. Are there any safety or injury risks that come with harvesting?
5. How many workers are usually needed to harvest one tree or a section of the farm?
6. Have you ever tried or seen harvesting machines or tools before? If yes, what did you like or dislike about them?
7. What features would be most valuable to you in a harvesting solution?
8. What kind of terrain and tree height should the solution be able to handle?
9. How much are you willing to invest or spend on technology that can make harvesting easier or faster?
10. Would you be open to testing or giving feedback on a prototype harvesting solution?



Data Extraction

Yasser Ashraf Aly

AI-based
disease
diagnosis

Machine
vision for fruit
counting

Remote
monitoring
dashboard for
farm status

Predictive model
for labor planning

Drone monitoring
for crop health

Drone monitoring
for crop health

Smart sensors to
detect fruit size &
color

Lightweight
harvesting robot
arm

Voice alerts for
overripe or
pest-affected
crops

Drone-assisted
mapping of ripe
areas



Data Extraction

ChewYJ

Equipped with
camera,
pruner, net

Polybag to
control pest
and disease

Harvesting,
sorting, and
grading

Challenges:
Labour &
whether

Automated
ripeness
detection

IoT system

Colour,
firmness, avg
fruit size

Open
fertigation
system

Whether
sensor trigger
raspberry pi

Starink
internet



Data Extraction

Vin Cent

Manually with
gloves,
clippers.

Labour
availability
and weather.

Manual
harvesting
and grading.

Falling from
ladders,
inexperienced
workers.

Fifty to sixty
worker per
farm.

Harvesting
machines
before not
cost effective

Automated
ripeness
detection,
AI-control

Flat but
sandy/muddy
terrain, ten
feet tall.

Open to
cost-effective
solutions

Visit farm for
testing



Data Extraction

Afifah

Better
control

Ripeness
based on
colours

Ripeness
based on
firmness

Start harvest
early in
morning

Ripeness
based on good
sizes

Better
cutter

Challenge
with labor

Challenge
with weather

Tracking of
planting (Hari
Selepas Tanam)

Open farm
system
(outside farm)

Harvesting
per hour

Need catcher or
net so it maintain
the shape

Data Clustering

AI and Automation	Machine Vision and Sensing	Drone and Remote Monitoring	Harvesting & Post-Harvest Systems	Farm Management & Infrastructure	Challenges and Constraints
- AI-based disease diagnosis - Automated ripeness detection (color, firmness, size) - AI control for harvesting timing - Predictive model for labor planning	- Machine vision for fruit counting - Smart sensors to detect fruit size & color - IoT system with weather sensors triggering Raspberry Pi - Parameters: color, firmness, avg fruit size	- Drone monitoring for crop health - Drone-assisted mapping of ripe areas - Remote monitoring dashboard for farm status - Starlink internet for stable connectivity	- Lightweight harvesting robot arm - Automated harvesting, sorting, and grading - Voice alerts for overripe or pest-affected crops - Catcher/net system to maintain fruit shape - Improved cutters - Polybag for pest and disease control - Harvesting rate tracking (per hour)	- Open fertigation system - Planting tracking (Hari Selepas Tanam) - Open farm system (outdoor environment) - On-site farm visits for testing and validation	- Labour availability (50–60 workers per farm) - Manual harvesting and grading - Injuries (falling from ladders, inexperience) - Weather dependency (rain, heat) - Flat but sandy/muddy terrain - Past harvesting machines not cost-effective - Early morning harvesting for quality

Personification



Profile

- Pak Rahman – The Experienced Farmer
- Age: 55
- Owns a medium-sized fruit farm (5 acres)
- Has 30 years of farming experience
- Uses traditional methods, limited tech knowledge

Pain Points

- Labor shortage and high worker turnover
- Unpredictable weather often disrupts his harvest
- Difficulty identifying ripe fruits accurately

Needs

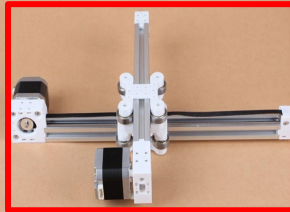
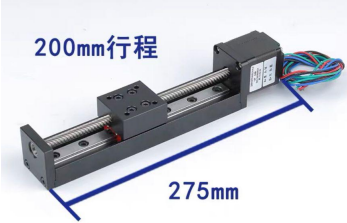
- Easy-to-use automated tools (plug-and-play)
- Reliable weather and ripeness alerts
- Affordable harvesting solutions
- Local support and training for new technologies



Design Statement

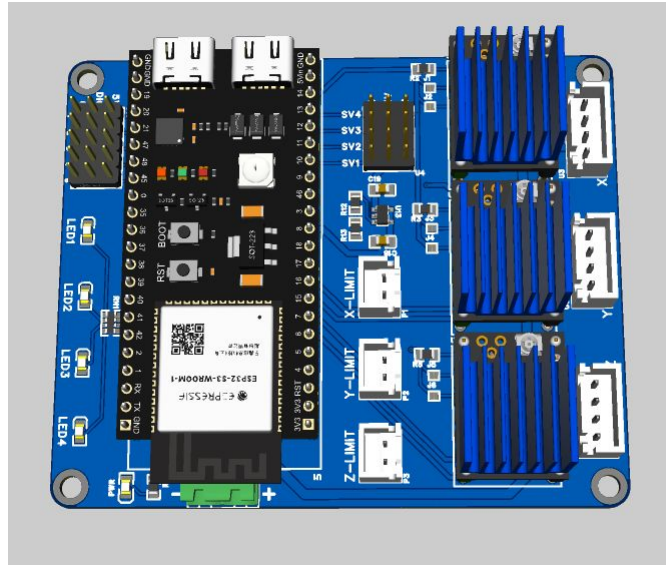
How might we help **Pak Rahman** to **harvest effectively despite labor** by **providing efficient, consistent, affordable automation tools** that adapt to his existing workflow?

Solution

	Option 1	Option 2	Option 3	Option 4
Picking Mechanism				
Movement				
Camera				
Storage				

Conceptual Sketch

XYZ axis automated
lemon harvesting
prototype



Stepper motor

Webcam

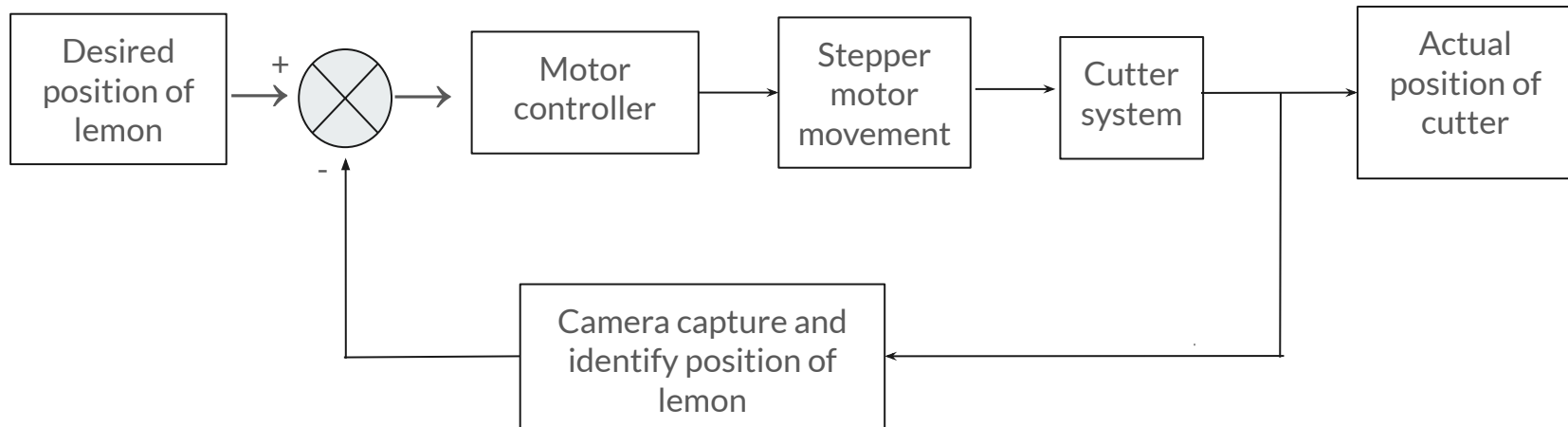
Servo controlled
cutter

Net to catch
falling lemons

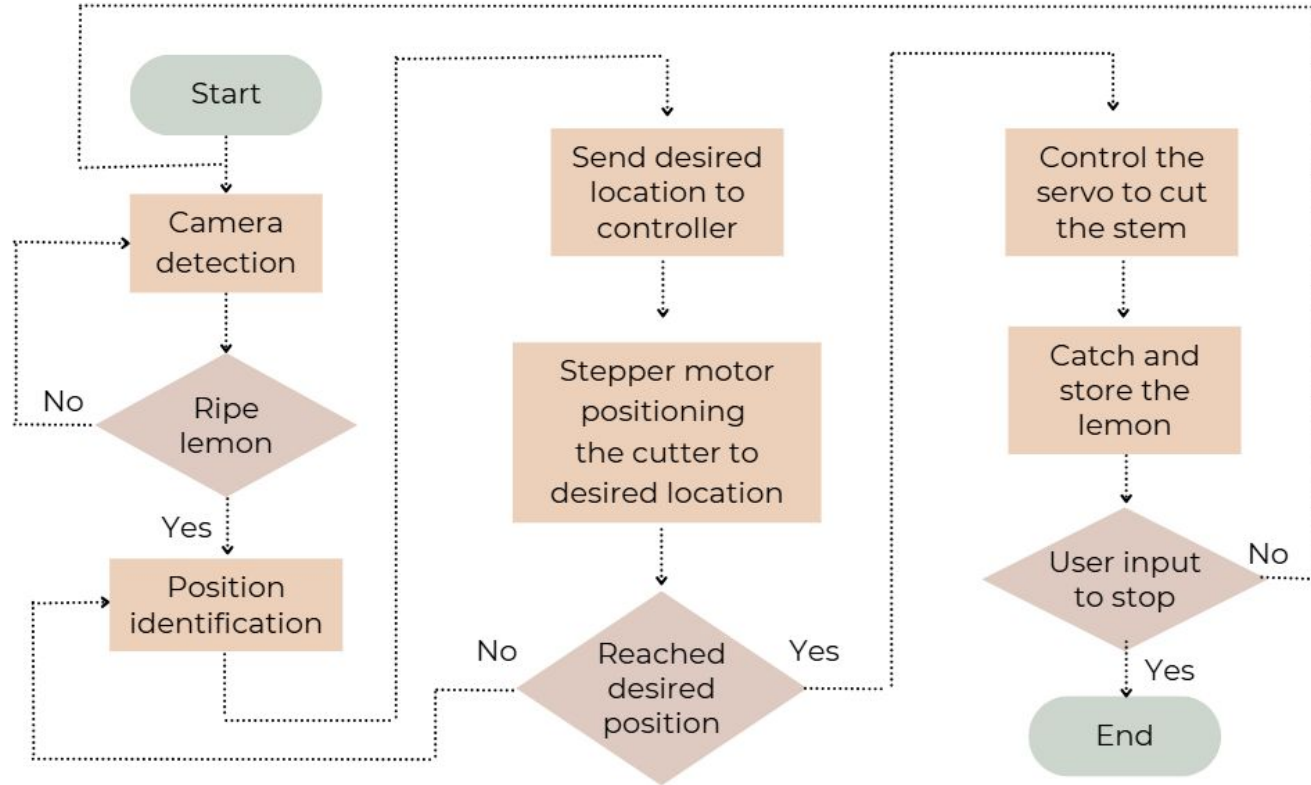
Conveyor belt
motion control

Caster
wheel

Conceptual Block Diagram



Conceptual Flow Chart



Comparison of Product

Product				Proposed lemon harvesting robot
Harvesting mechanism	Grip & Pluck	Pick & Place	Suction cup (vacuum gripper)	Hook & Cut
Picking accuracy	95%	85%	90%	80%
Fruit damage rate	50%	20%	30%	10%
Mobility	4 DOF	Semi-mobile	Autonomous	Fixed with XYZ movement
Cost	RM 10,000	RM 40,000	RM 600,000	RM300

Expected Cost

Product Cost (Week 1-4)					
No	Planning Components. <i>Only the componeed need in the product is considered here</i>	Link / Specification	Unit Price (RM) <i>(Don't include backup quantity in the list)</i>	Quantity	Subtotal (RM)
1	Linear Slide Module Synchronous Belt Slide Table 42 S	https://click.world	103.55	1	103.55
2	Gantry synchronous belt slide table aluminum profile 20	https://item.taoba	52.81	1	52.81
3	Servo MG90s		8.00	1	8.00
4	Thread U Shape Sissors	https://item.taoba	0.58	1	0.58
5	USB Web Cam		30.00	1	30.00
6	Custom Made Motor Controller with ESP32		80.00	1	80.00
7	3D Print Parts		20.00	1	20.00
8	Power Supply 24V (loan)			1	0.00
TOTAL PRODUCT COST					294.94

S.T.E.E.P Analysis



Social

- Helps older farmers work more safely and easily.
- Reduces the need for many workers on the farm.

Technology

- Uses simple automation that's easy to learn.
- Includes sensors and cameras for accurate fruit picking.

Economic

- Lowers labor and operation costs.
- Increases profit by improving harvest efficiency.

Environmental

- Reduces fruit waste by picking at the right time.
- Supports eco-friendly and careful farming methods.

Political/Legal

- Matches government goals for smart farming.
- May qualify for farming grants or support programs.



Thank You